

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
LABORATORY OBSERVATION

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Branch & Sec: CSM-B

Date of Execution: 02 / 08 / 2026

Course Name: Full Stack Web Development

Course Code: 23CM4121

EXPERIMENT NO: 3

1. TITLE OF THE EXPERIMENT:

Implementation of JavaScript ES6 Classes, DOM Manipulation, and Dynamic Form Binding

2. MAPPED OUTCOMES:

- **Course Outcome (CO):** CO1
- **Program Outcomes (POs):** PO1, PO2, PO3, PO4, PO5

3. AIM:

To design and develop a responsive "Student Portal" web application demonstrating ES6 Class concepts, form event handling, and dynamic DOM manipulation for data capture and display.

4. ALGORITHM / DESIGN:

Algorithm: Design of Student Portal

Step 1: Document Structure & Styling — Initialize the HTML5 document containing two main layout sections inside a Flexbox container: an input form card and an output display card. Style components using clean UI modern CSS, rounded cards, custom borders, and responsive media queries.

Step 2: Define ES6 Class Schema — Create a JavaScript `Student` class with a constructor accepting `name`, `rn`, `dept`, and `cgpa` to encapsulate student entity data.

Step 3: Data Extraction & Instantiation — Define the `createstu()` function to extract user inputs from DOM elements (`iname`, `irn`, `idept`, `icgpa`) by their IDs and return a new instance of `Student`.

Step 4: DOM Binding & Display — Implement the `display(stu)` function to dynamically update target output placeholder elements (`uname`, `urn`, `udept`, `ucgpa`) with the property values of the passed `Student` instance using `innerText`.

Step 5: Event Integration & Prevention — Attach the inline event handler `onclick="display(createstu())"` to the submit button while setting `onsubmit="event.preventDefault();"` on the form to avoid default page reload behavior.

5. TEST CASES & EXECUTION DATA:

Test Cases (Validation & UI Verification)

- **TC-01:** Input: Load the webpage in a modern web browser
Expected : UI renders two centered cards ("Enter Student Details" form and "Student Details" placeholder card) side-by-side | Status: Passed
 - **TC-02:** Input: Enter Full Name: "Ramu", Roll Number: "101", Department: "Computer Science", CGPA: "8.9", and click Submit
Expected : Right card updates dynamically displaying Name: "Ramu", Roll No: "101", Department: "Computer Science", CGPA: "8.9" without page reloading | Status: Passed
 - **TC-03:** Input: Enter decimal values in CGPA field (e.g., 9.25) within step range (0 - 10)
Expected: CGPA input accepts valid step and bounds accurately | Status: Passed
 - **TC-04:** Input: Hover and focus over input fields and action button
Expected: Input outline highlights with border-color transition to #0066cc; button transitions background smoothly to #333333 | Status: Passed
 - **TC-05:** Input: Resize browser viewport to screen width < 600px
Expected: Container switches from row-direction flex to single-column vertical layout smoothly | Status: Passed
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6. OUTPUT PROOF:

The image shows two side-by-side cards on a light gray background. The left card, titled 'Enter Student Details', contains four input fields: 'FULL NAME' with the value 'Ravi', 'ROLL NUMBER' with '95', 'DEPARTMENT' with 'CSM', and 'CGPA' with '9.1'. Below these fields is a black 'Submit' button. The right card, titled 'Student Details', displays the entered information: 'NAME: Ravi', 'ROLL NO: 95', 'DEPARTMENT: CSM', and 'CGPA: 9.1'.

Github Link:

https://github.com/purnachandra1052006-collab/FullStack/tree/main/Observation/Task_3

7. LEARNING OUTCOME / RESULT: Successfully created and tested the Student Portal application. Gained hands-on experience in implementing Object-Oriented Programming principles in JavaScript using ES6 classes, reading form input values via DOM element selectors, preventing default form submission behaviors, and updating UI elements dynamically at runtime.

RUBRIC EVALUATION (Faculty Use Only)

1. **Design & Logic** (5 Marks):
2. **Execution & Output** (5 Marks):
3. **Viva & Record** (5 Marks):

TOTAL MARKS: [/ 15]

Faculty Signature: _____